N-R-002189(Army-GL)
October 6, 1983

INTERIM FEDERAL SPECIFICATION

RICE, WHITE, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by the U.S. Army Natick Research and Development Laboratories, Natick, MA 01760, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers white rice thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, and other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

FSC 8920

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Military Specifications:

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items
MIL-L-35078	-	Loads, Unit: Preparation of Nonperishable Subsistence Items; General Specifications for

Military Standards:

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-900	-	Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Canned Foods for the Armed Forces.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1 - 199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture (USDA)

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Food Quality Assurance Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

United States Standards for Milled Rice

(Application for copies should be addressed to the Director, Fruits and Vegetables Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Grades of Butter

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

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Environmental Protection Agency (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

National Research Council

Food Chemicals Codex

(Application for copies should be addressed to the Office of Director, National Academy of Sciences, National Research Council, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.1 and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Rice, white, parboiled. Parboiled long grain rice shall be U.S. No. 1 and of such varieties as Belle Patna, Bluebelle, Bluebonnet and Lebonnet or similar varieties. The parboiled rice shall contain not more than 2 percent of kernels having white ungelatinized areas and not more than 14.5 percent moisture.

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3.2.2 Butter, lightly salted. The lightly salted butter shall be U.S. Grade A or better as defined in the U.S. Standards for Grades of Butter.

3.2.3 Glucono delta lactone. Glucono delta lactone shall comply with the requirements of the Food Chemicals Codex.

3.2.4 Salt. Salt shall be white, refined, sodium chloride with or without anticaking agents. Iodized salt shall not be used.

3.2.5 Sugar. Sugar shall be white, refined, granulated, cane or beet sugar, or a combination thereof. Sugar shall meet the bacterial requirements of MIL-STD-900.

3.2.6 Water. Water used for formulation and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.3 Preparation and processing.

3.3.1 Rice preparation. The dry rice shall be thoroughly blended with melted butter at the level of 3 ounces butter per pound of rice.

3.3.2 Brine preparation. The brine shall be prepared by dissolving the following ingredients into each gallon of water:

a. Salt - 3.00 oz
b. Sugar - 6.00 oz
c. Glucono delta lactone - 1.20 oz or quantity sufficient to meet the pH requirement of paragraph 3.6

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with 40 ± 1 ounces of rice and butter mixture and approximately 66 ounces of brine such as to conform to the finished product requirements listed below. Each can shall be hermetically sealed under a vacuum of 15 to 18 inches immediately after filling.

a. The average net weight shall be 106 ounces minimum.
b. No individual can shall contain less than 104 ounces of product.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting at 220°F until the internal product temperature of the coldest can reaches 200°F.

NOTE: Extreme caution shall be taken not to overprocess this product.

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3.6 Finished product. The finished product shall be off-white, slighty starchy, rice grains with a moderately acidic flavor. The finished product shall be soft and fluffy after thorough stirring, shall remain as distinct particles, and shall not form a gummy mass. The finished equilibrium pH of the rice shall be no higher than 4.55 and not less than 4.25. The moisture content of the finished product shall be 58 ± 2 percent.

3.6.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed and packaged under sanitary conditions in accordance with good manufacturing practice. The product shall be processed in establishments meeting the sanitary requirements of the applicable government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this specification. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1 Ingredient examination. Conformance of ingredients to identity and condition requirements shall be determined by examination of pertinent labels, markings, US Grade Certificates, invoices, certificates of analysis, or other

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valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient lot or of any involved product.

4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1.0 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

4.6.2 Product inspection. Eight tray packs shall be selected at random from the lot. The eight tray packs shall be heated at least 15 minutes in 140° to 150°F water, opened, and inspected in accordance with table I.

a. If no defects are found, the lot shall be acceptable.
b. It two or more defects are found, the lot shall be rejected.
c. If only one defect is found, a second sample of eight tray packs shall be selected at random from the lot, heated as above, and inspected for the defects listed in table I. If any defect is found in the second sample, the lot shall be rejected.

TABLE I. Product inspection 1/ 2/

Category	Defect
Major	
151	Color not off-white to cream colored (materially darkened, scorched, or oxidized).
152	Texture of rice hard, mushy, or pasty.

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TABLE I. Product inspection 1/ 2/ (cont'd)

Category	Defect
Major	
153	Appearance not slightly fluffy, distinct, uniform rice grains.

1/ The presence of any foreign material (e.g., glass, dirt, insect parts, hair, wood, metal, etc.), foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale, etc.), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

4.6.3 pH testing. Three tray packs shall be randomly drawn from each lot. For each sample tray, three aliquots shall be randomly drawn from different parts of the tray and the pH determined in accordance with the *Cereal Foods* Chapter of the Official Methods of Analysis of the Association of Official Analytical Chemists. The results shall be reported to the nearest 0.01. The average of these three determinations will constitute the pH of the unit sample. The average of the lot shall be calculated using these unit sample determinations. Acceptance of the lot shall be based on the lot average pH being between 4.25 and 4.55 and no unit sample pH used in determining the lot average being less than 4.20 or greater than 4.60.

4.6.4 Moisture content testing. Three tray packs shall be selected at random from the lot and individually tested for moisture content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Fruits and Fruit Products, Method: Moisture in Dried Fruits. The results shall be reported to the nearest 1 percent. Any test result falling below 56 percent or exceeding 60 percent shall be cause for rejection of the lot.

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497.

4.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement shall be in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: National stock number, item description,

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contract number, or date of pack markings missing, incorrect, or illegible. Minor: (1) Other required markings missing, incorrect, or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) Reinforcement with nonmetallic strapping or tape is not used, (4) Height of box liner less than inside depth dimension of box by more than 1/8 inch or extends above top edge of box.

4.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 106 ounces of the food product, sealed and thermoprocessed as specified in 3.4 and 3.5. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except the cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the fill inside depth of the box (+0 inch, -1/8 inch). The pads shall be full length and width and fabricated of grade V3c fiberboard. The pads shall be full length and width less approximately 1/4 inch in each dimension. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s or V4s fiberboard.

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5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be full length and width and be fabricated of the same material as the box. The pads shall be full length and width and be fabricated of the same material as the box.

5.3 Unit loads. When specified (see 6.1), the product packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each package shall be labeled in accordance with MIL-L-1497. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT

a. To heat in water: Submerge unopened can in boiling water. Simmer gently 35 minutes. Remove from water and open lid for serving.

b. To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 35 minutes until contents reach an internal temperature of 165°F.

WARNING

Do not place closed can in oven as heating will increase internal pressure which may cause the can to burst.

YIELD

18 servings approximately 1/2 cup each.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Provisions for approved preproduction samples (see 3.7.1 and 6.3).
d. Level of packaging and packing required (see 5.1 and 5.2).
e. Type and class of unit load when unit loading is required (see 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encoutered during shipment, handling and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article shall be a preproduction sample. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for selection, inspection, and approval of the first article.

Preparing activity:

Army - GL

Project No. 8920-A441

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
N-R-002189(Army-GL)	Rice, White, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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